Science

Nature Walks & Scouting
General Science
Labs
Natural History
Nature Notebook

Alveary

Science: Grade 5

GRADE(S): 5

About the Course

In level 5, learners expand their knowledge of science in their world and continue building personal connections as they discover a variety of topics from botany, astronomy, and more.

This course includes the following topic(s): General Science: Grade 5, Natural History: Grade 5, Labs: Grade 5, Nature Notebook: Grade 5, Nature Walks & Scouting: Grades 1-8

About Nature Walks & Scouting: Grades 1-8
Students spend time once each week on a longer excursion. Prompts can be found in Outdoor Work, if desired. This digital resource is provided with all science lessons.

About General Science: Grade 5
Learners engage with the chemical nature of Things, meteorology, space exploration, and robotics as they practice beginning lab skills and expand their use of graphs. The lab component is required for this course. Coordinating afternoon activities are provided in Outdoor Work.

About Labs: Grade 5
This is the required lab component for level 5 General Science. This course includes our custom laboratory book which will build foundational and progressive skills and habits.

About Natural History: Grade 5
Learners encounter botanical families, endangered species, invasive species, and secondary succession as they consider how man interacts with and thinks about the natural world. Coordinating afternoon activities are provided in Outdoor Work.

About Nature Notebook: Grade 5
Students spend time at least once each week recording their observations. Prompts that coordinate with curricular content can be found in Outdoor Work, if desired. This digital resource is provided with all science lessons.

Science: Grade 5
The Big Picture:
To accomplish the goal of supporting a relationship with the Things of the Universe, a Mason science program consists of nature lore, natural history, and general science. Nature immersion, inquiry, community connection, and supportive literature are woven into each of these three parts. Building on the familiarity and foundational knowledge of Form 1, Form 2 learners in grades 4-6 extend the scope of the Things they encounter and begin to notice that science is active in their world. Growing in their self-knowledge at this age, they may notice that they gravitate toward some of these Things more than others. This is a bit like noticing that you have a natural rapport with a particular friend because they seem to understand the way you think and share the same interests. This is wonderful and very typical, but we still want them to get to know more “friends” in science and try different Things, even though one might be their particular interest. Their sense of community is expanding, as well: who are the scientists in their world, and what are they doing? Special care has been taken to build vocabulary and conceptual knowledge “by the way” and to maintain the connection between scientific endeavors and the human experience. In Form 3, students will continue to build on

this knowledge of what science is doing in the world when they examine science in its historical context.

Nature lore is timeless knowledge that is passed through a community, much like a grandmother passes on how to make that special bread when the dough just "feels right." Like Mason, we strive to pass on this knowledge primarily through outdoor work. Group nature walks, seasonal readings, and topics in scouting are provided as an Outdoor Work resource in the Quick Links. If desired, literature suggestions to support lore can be found in the Community Read Alouds resource (in Citizenship Grades 4+).

Placement & Combining Tips

Science: Grade 5
For fifth-grade students or possibly hungry fourth-graders or sixth-graders taking their time. If combining grades is desirable: the combined grades may begin at the appropriate level and move forward together, learners may stay at their appropriate level for General Science while combining Natural History and Outdoor Work, or teachers can adapt the laboratory content and reading level to meet their needs.

Nature Walks & Scouting: Grades 1-8
Learners may be combined and share prompts or follow the plan of their local scouting troop or natural history club. Support accessibility and sensory development as appropriate.

General Science: Grade 5
For fifth-grade students or possibly hungry fourth-graders or sixth-graders taking their time. If combining grades is desirable: the combined grades may begin at the appropriate level and move forward together, or teachers can adapt the reading level and laboratory content to meet their needs.

Labs: Grade 5
The ideas and skills in this component are progressive, like math or grammar. Teachers should read the lab book thoroughly to understand what concepts might need to be supplemented should they choose a different sequence or a substitution.

Natural History: Grade 5
For approximately fifth-grade students based on a balance of complexity and reading level. However, learners may be freely combined, or sequence reordered based on needs and preferences.

Nature Notebook: Grade 5
Learners may be combined and share prompts or follow their own interests. Support through varied media, scribing observations, or notebooking in tandem.

Scheduling

GRADE	SCHEDULE INFO.	BOOKS
1-8	Nature Walks & Scouting: Grades 1-8 1 time/week 30 min+	
5	General Science: Grade 5 1 time/week 20 min	Mission to Pluto: The First Visit to an Ice Dwarf and the Kuiper Belt The Hive Detectives: Chronicle of a Honey Bee Catastrophe The Tornado Scientist
5	Labs: Grade 5 1 time/week 30 min	Science: Grade 5 Lab Book
5	Natural History: Grade 5 1 time/week 15 min	Beetle Busters Nature All Around: Plants Shanleya's Quest Card Game

GRADE	SCHEDULE INFO.	BOOKS
		The Bat Scientists The Forest in the Tree
5	Nature Notebook: Grade 5 1+ time/week 15 min+	

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Grade 5				
Natural History: Grade 5		General Science: Grade 5 Nature Walks & Scouting: Grades 1-8		Labs: Grade 5 Nature Notebook: Grade 5

Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Science: Grade 5
☐ Obtain any supplies indicated on the science or grade-level supply lists.
☐ Download any apps and shortcut any desired links.

Nature Lore:
☐ Bookmark your Outdoor Work Quick Link, so that you have it available on your weekly outing. Outdoor Work is generally flexible for your location and season and can be moved around in the schedule to incorporate or substitute Natural History Club outings.
☐ Print or bookmark grade-specific nature notebook suggestions to support natural history and general science. Notebooking can be done on a walk, during occupations, or as a field trip, as appropriate.

Special Topics & Field Trips

Science: Grade 5
Encourage students to begin choosing their own special topic to observe, study, and research in their field guides. What or who are they curious about or interested in getting to know better? Teachers can choose their own special study, too! Suggested field trips include:

General Science: Grade 5
☐ Term 1: a beekeeper
☐ Term 2: a meteorologist or weather station
☐ Term 3: evening walks, a planetarium

Natural History: Grade 5
☐ Term 1: a botanical garden or plant conservatory
☐ Term 2: evening walks and/or a cave, a wildlife rehabilitator, or bat conservationist
☐ Term 3: a forestry museum or tour

Term Prep & Teacher Tips

Science: Grade 5
Object lesson prompts are woven into the lessons, as appropriate, but teachers are encouraged to read corresponding Handbook of Nature Study (HoNS) selections each term and/or any that support special topics chosen by students:

General Science: Grade 5
☐ Term 1: bees p.384-395
☐ Term 2: climate and weather p.780-783, the winds and storms p.791-799
☐ Term 3: stars p.815-821

☐ Gather household items, typically easy for students to scavenge or teachers to obtain locally:
- honey
- food items suggested
- spoons
- small saucepan
- plant fertilizer of your choice (only a small amount)
- water
- flowers (Term 1)
- a strainer or sieve
- 1 c dry soil
- kitchen scale
- kitchen thermometer, any stick-type tool that can be placed into hot water and soil
- oven
- oven mitt
- clock
- chair or stepladder
- stopwatch, such as found on most electronic devices
- freezer
- stove
- 1-2 c white vinegar
- 1/2-1 c baking soda
- flat surface outside to launch rocket
- 10-20' height reference, such as the side of a building, a long piece of wood, or a flagpole
- various supplies by student design
- printables
- optional: barometer
- optional: computer

Natural History: Grade 5
☐ Term 1: one or more wildflowers from p.461-508
☐ Term 2: bats p.241-244
☐ Term 3: one or more trees from p.628-692, one or more insects from p.301-395

☐ Gather household items, typically easy for students to scavenge or teachers to obtain locally:
- flower pot with soil, or place in the ground
- trowel
- globe/wall map

Reminders

General Science: Grade 5
☐ Note that Lab in week 1 begins with 5 "unknown" food substances (1- 2 Tbsp) to include honey and any other various items, e.g. peanut butter, whole grain bread, white bread, raw egg white, cereal, applesauce, jelly, whole milk, milk substitute, juice, flavored yogurt, plain yogurt, etc. These will be provided to students in 5 similar-looking containers for study. Refer to the lab

book for more details.

Natural History: Grade 5

☐ Note that term 1 of Natural History begins with planting a small garden, so be sure to have those supplies and any others indicated on the supply list ready. For indoor situations, the choices are many, provided there is a sunny window. For outdoors in a cooler season, some good options are parsley, carrots, or peas (may need a trellis). Basil, thyme, or beans (may need a trellis) are good options for outdoor planting in a warmer season. Note that some plants will not reach the flowering stage for a while.

Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ View Book List Details

Science: Grade 5

General Science: Grade 5

 Mission to Pluto: The First Visit to an Ice Dwarf and the Kuiper Belt

 The Hive Detectives: Chronicle of a Honey Bee Catastrophe

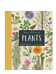
 The Tornado Scientist

Labs: Grade 5

 Science: Grade 5 Lab Book

Natural History: Grade 5

 Beetle Busters

 Nature All Around: Plants

 Shanleya's Quest Card Game

 The Bat Scientists

 The Forest in the Tree

Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ View Basic Supplies

∞ View Supply List Details

Science: Grade 5

Alveary Grade 5 Science Kit

 48 Prepared Microscope Slides for Beginner

Nature Walks & Scouting: Grades 1-8

- Wild Bird Seed
- Bird Feeder
- Hand Lens
- Small Collection Containers
- Small Basin
- Dip Net
- Student Microscope

General Science: Grade 5

- Household Items - General Science: Grade 5

Labs: Grade 5

- Small Collection Containers
- Slides and Coverslips
- Safety Goggles
- Scale/Balance
- Robot-Building Kit
- RC Car
- Student Microscope

Natural History: Grade 5

- Household Items - Natural History: Grade 5

Quick Links

Click THIS text or scan the QR code for links.

Science: Grade 5

- Outdoor Work
- Seek app from iNaturalist
- SkyView Lite for iOS
- SkyView Lite for Android

Labs: Grade 5

- Grade 5 Lab Book
- Lab Notebook Examples

Science: Grade 5

How To Teach

Introduce

• Begin each lesson with learners' existing knowledge. If the book or activity is new or unfamiliar, then look at the title, a picture, or guidance in the lesson plan to help discuss what students think, drawing on previous experience. If continuing or revisiting a topic or activity, then recap.

• Often, the introduction in the lesson plans or the title of the book's section can help learners to draw out the main idea. Use this to support learners, as appropriate.

• Some learners may benefit from using pictures or looking back briefly.

• Allow them time to share any concerns and come alongside, as needed.

Read

• Read or do, as instructed in the lessons, noting any Teacher Tips provided. Learners should always have their journal, notebook, or other paper available in case they need to draw or diagram during the lesson.

• Use supportive strategies and educational tools to reduce frustration and better engage the mind, as appropriate. These could include, but are not limited to, the use of eBooks, pictures, audio, read-aloud, buddy reading, colored reading strips, etc.

• If learners do not understand a word or concept, do not worry. Try to show them with a picture or connect the idea to something they have seen in real life. They are learning much by the way and will likely build understanding over the term, the year, and beyond.

Narrate

• Process the ideas of the lesson by retelling events, describing, explaining a concept, etc. Tips about helping learners deal with various types of passages are provided, but teachers can learn more in Charlotte Mason's School Education Chapter 16, especially p.180.

• Learners may use words, pictures, Legos, PowerPoint, etc, to process and convey ideas in their own way.

• Teachers may take turns to model.

Discuss

• Consider together any thoughts, confusion, or concerns about the passage, keeping in mind that oftentimes new concepts are not going to be 'mastered' on the first or even second introduction. Mastery is not necessarily the goal, but curiosity and thinking.

• Questions/topics for further discussion are often provided in the lesson plans (or even lab books) to help. There are no right or wrong answers to these. Alternatively, many of these can be used for composition, depending on the needs of the learner and the instructional goals of the teacher. If teachers want to keep ideas active in the mind, these can also be used at other times to keep the ideas in the working memory.

- Note that learners may need to spend more than the allotted time engaging with or even repeating a lesson before moving on or as reinforcement at a later time. Adjust the pace as needed to feed the learner.

- Notice if there were any dates that they want to keep for their Book of Centuries.

Connect

- Follow any extra links, examine any sidebars in the text, look at pictures, etc., depending on learner needs and interest.

- These can be viewed as alternative ways to engage.

Science: Grade 5

How To Teach

Introduce

Regardless of how many days are required to complete a particular activity, every science lab has the same flow, which follows the scientific method and is guided by the lab book.

• On day 1, learners read an introduction in their lab book. How does this lab activity relate to what they have learned so far, as well as any previous experience? What will they learn from the lab? This is analogous to the conversation we might have when we begin something new in any subject, but they may need to dialogue as they learn to extend these skills to the laboratory.

• Once they have had a chance to think, they will compose a prelab narration to put these introductory ideas into their notebooks. The prompt in the lab is generalized and consistent, so they learn the habit. Eventually, they will learn to formulate this as a hypothesis. For example, a learner preparing for a lab about the use of insect repellent might write:

"I have read about some diseases that are spread by insects, like Lyme disease. I also know that my sister is allergic to some insect bites. Insect repellent contains pesticides to keep insects away. Some scientists worry about how pesticides affect wildlife. I am going to compare some different insect repellants in this lab to see if they really work."

• Written narration and composition are skills that they will build over time. These prelab narrations may seem short and even incomplete at first, but that is okay. If learners have difficulty or are easily frustrated, then provide them with support. Teachers may act as scribes or allow students to keep a digital notebook to type or use assistive technology, as appropriate.

• After they complete their prelab narration, the listed materials are collected. This gives the learner some responsibility to let the teacher know if something is missing or to remind the teacher if something needs to be purchased at the store.

• If these activities on day 1 do not fill the scheduled time, that is fine. They might use the additional time to familiarize themselves with the procedure, draw a picture from their book, or catch up on any other work. Some labs may instruct the student to begin on the same day.

Lab Procedure

• Then, students begin and follow the procedure (whenever prompted by the lab book). Note that the lesson plans guide teachers as to which sections are completed each week, and the lab book instructs learners when to take a break.

• The lab gives instructions for using their notebooks to create tables and figures, as needed. Do not allow this to become an obstacle. Do it with them, first having them watch and then having them copy or help when ready.

• If teachers choose to have learners record directly in the lab book or on a photocopy, then cut out and tape these into their lab books, so that they can see how the record is built.

• Learners may feel unsatisfied with their results. This is often part of the process. Come alongside, help them learn to be okay with uncertainty and questions, and teach them what to do with it in the next step.

Analysis & Conclusions

● The last step in the lab is to analyze the data and observations and draw conclusions from them. For some simpler labs, learners will complete their analysis and concluding (or postlab) narration on the last day of the lab procedure. For labs that are more involved, a separate day has been built in to allow adequate time for this.

● Similar to the prelab narration, the concluding narration is a chance to think about and put into words (or questions). This time, they are considering what they learned from the lab, what they could learn more about if they were to continue, and possibly how they would pursue that learning. Again, they might need support in the form of dialogue, a scribe, etc. For example, the above learner might write:

“It was clear that the Off and black pepper essential oil worked against ants because they would not even touch the line of repellent, but I wasn’t sure about the Skin So Soft because it ran all over the place. I could test this one again with different insects or on a different surface.”

● Depending on the interest of the learner and the priorities of the teacher, the student might be encouraged to spend more time on those ideas of what more they could learn, or it might be time to move on. Either way, it is an important part of the scientific method to reflect on what we could or would do next - our practice should help to clarify our thinking and teach that there is always more to be learned.

● Teachers should engage learners with this reflection by reviewing their lab notebooks with them, discussing the science used in the lab, and demonstrating curiosity about the lab themselves.

Science: Grade 5

Click THIS text or scan the QR code for links.

Term 1

WEEK 1

⬚ **15m Natural History: Grade 5 - Lesson 1**

My Own Garden

▯ Materials: Nature All Around: Plants, video links, seeds, pot with soil or place in the ground, trowel, water

PREP: Read Teacher Tip

→ INTRO

Did you ever think of all the world as a garden full of many different plants? What about plants stands out to you most? Look at the picture on the cover of this book. How many different plants do you see? Do you recognize any of them?

→ READ, NARRATE, & DISCUSS

p.4-6 "Do you have" - "four basic parts." Take a moment to diagram a plant with these 4 parts.

p.8 "Flowering plants start" - "They shrivel up."

→ VIEW

Let's watch the beginning of the plant life cycle, and then we'll begin our own gardens, choose one or both based on time:

∞ Video Link: Growing Basil Time Lapse (1:42)

∞ Video Link: Wildflower Garden Time Lapse (3:31)

→ ACTIVITY

Begin your own small garden in the ground or even in a single pot. Plant seeds according to the seed packet and plan to check on them and water them regularly. The soil should be moist, but not wet. If it seems wet, then don't add more water. If you gently blow across the surface and the soil has a dusty character, then give it some water. As seeds sprout over the next few weeks, make careful observations in your nature journal.

→ NARRATE & DISCUSS

Use a nature journal to record what you planted.

★ TEACHER TIP

The livingness of this book lies in its artwork, so read the assigned pages, but know that some students may learn more from reading only a part and instead exploring and spending time with the pictures. When students narrate, they are analyzing the "big idea."

● OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 1

⬚ **20m General Science: Grade 5 - Lesson 1**

Visit a Hive

▯ Materials: The Hive Detectives

PREP: Read Teacher Tip

→ INTRO

Look at the cover picture and the main title. Have you watched bees before? What did you notice about them? What would you investigate if you were a "hive detective"? Read the subtitle. Do you know what a chronicle is? (a factual record) Do you know what a catastrophe is? (a great disaster) So, this is a story about scientists who are studying a disaster in a beehive to learn how it happened in the hopes that they can prevent it from happening again.

★ TEACHER TIP

Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to the discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and

Science: Grade 5

Click THIS text or scan the QR code for links.

Term 1

	→ READ, NARRATE, & DISCUSS p.1-3 "Put on your" - "home in Wisconsin." **→ SUPPLEMENTAL** ∞ Video Link: Why do Honeybees Love Hexagons? (3:58) ∞ Video Link: Why Do Bees Build Hexagons? (2:34)	see what works best for your student(s).
WEEK 1	⬚ **30m Labs: Grade 5 - Lesson 1** *Lab: What is Honey?* Materials: Grade 5 Lab Book and materials listed within PREP: Read Teacher Tip **→ LAB DAY** Complete day 1 of What is Honey?, as directed in the Lab Book. Suggested day 1: Students read the Introduction and then compose the prelab narration in their lab notebook. These need not be more than 1-3 sentences at first, and teachers should feel free to scribe for students, as necessary. Remaining time can be spent gathering materials and beginning steps 1-2 of the Procedure.	★ TEACHER TIP Pacing is only a suggestion. Students should engage with the lab at a pace that is appropriate for their abilities and interests.
WEEK 2	⬚ **15m Natural History: Grade 5 - Lesson 2** *We Depend on Plants* Materials: Nature All Around: Plants, video links **→ INTRO** What do you recall from our last reading about plants? Why do you think plants are important to us? Plants make the air we breathe; they shape our communities and traditions; they even make many of our most important medicines. Let's find out how they do the first of these in our book. **→ READ, NARRATE, & DISCUSS** p.9 "A plant's green leaves" - "even at night." ●Add to last week's diagram or sketch a new one, showing how the plant releases oxygen into the air. **→ VIEW** ∞ Video Link: Ethnobotanist (to 3:03) ∞ Video Link: Life-saving Medicines (to 4:13) **→ NARRATE & DISCUSS**	● **OUTDOOR WORK** Weekly walks and daily outdoor time are vital! Resources in Quick Links.

Science: Grade 5

[Click THIS text or scan the QR code for links.]

Term 1

WEEK 2

⬚ **20m General Science: Grade 5 - Lesson 2**

Know the Bees

▯ Materials: The Hive Detectives

→ INTRO

What did you read last time?

→ READ, NARRATE, & DISCUSS

p.4-9 "Mary begins" - "for their keeper."

• Describe the structure of the hive and the jobs of the bees. Draw or diagram, if helpful.

WEEK 2

⬚ **30m Labs: Grade 5 - Lesson 2**

Lab: What is Honey?

▯ Materials: Grade 5 Lab Book and materials listed within

→ LAB DAY

Complete day 2 of What is Honey?, as directed in the Lab Book.

Suggested day 2: Students continue the Procedure today. They may draw, diagram, or dictate in their lab notebook what they observe, just as they would in their nature journals. Finish whatever is comfortable and enjoyable today. Do not allow the testing to become tedious. Help them use the grid lines in their notebook to make tables. If not ready to do so, they may cut and paste, as needed.

★ TEACHER NOTE

Be sure to determine local guidelines for chemical waste disposal.

WEEK 3

⬚ **15m Natural History: Grade 5 - Lesson 3**

Getting to Know Flowering Plants

▯ Materials: Nature All Around: Plants, video link, Shanleya's Quest card game

→ INTRO

Recall what we learned in the last lesson. What do we call someone who studies plants? What do they actually do?

→ VIEW

∞ Video Link: What is a Botanist? (6:31)

→ READ, NARRATE, & DISCUSS

p.10-11 "Take a look" - "from the flowers."

• If you have the Shanleya's Quest card game, then explore the deck today. Notice differences in the 8 plant families presented and how the key characteristics are repeated in different plants in the family. If you do not have the card game, then refer to the link (about halfway down the webpage).
∞ Website Link: Shanleya's Quest Patterns in Plants

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

Science: Grade 5

[Click THIS text or scan the QR code for links.](#)

Term 1

WEEK 3

⬚ **20m General Science: Grade 5 - Lesson 3**

Beekeeping Cross-Country

▯ Materials: The Hive Detectives

→ INTRO
What did you read last time? Some beekeepers manage thousands of bees. Who would need thousands of bees? Let's find out about what happened when one of these beekeepers visited his hives.

→ READ, NARRATE, & DISCUSS
p.11-14 "On November 12," - "a thankless job."

● Contrast how Mary and Dave keep their bees.
● What would it feel like to come upon the animals you care for to find them gone?

→ SUPPLEMENTAL
∞ Video Link: Honey Bee Shortage (2:34)

WEEK 3

⬚ **30m Labs: Grade 5 - Lesson 3**

Lab: What is Honey?

▯ Materials: Grade 5 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY
Complete day 3 of What is Honey?, as directed in the Lab Book.

Suggested day 3: Students continue the Procedure today.

★ TEACHER TIP
It is fine if students do not have time or interest for all of the tests. They can simply move on next week. There is time to spend another week on the chemical testing, either now or to come back to it at the end of the term, but this should be done based on student interest.

WEEK 4

⬚ **15m Natural History: Grade 5 - Lesson 4**

Caring for My Garden

▯ Materials: The Forest in the Tree, plantings from earlier in term

→ OBSERVE
Do your seedlings appear similar to or different from each other? Are some growing slower or faster than others? Are the seeds that you planted monocots or dicots? How do the different parts of the plant, like the roots, leaves, and flowers, serve its purpose? (Refer to Nature All Around: Plants, if you need help.)

→ NARRATE & DISCUSS
Use a nature journal to record what you observed. Do you think the soil that plants grow in is very important, or can they grow in anything? What do you think is in soil?

→ OBSERVE & DISCUSS
Look at the picture on p.34. Read all or parts of p.35 to accompany the picture.

● **OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

Science: Grade 5

[Click THIS text or scan the QR code for links.]

Term 1

• Compare the relationship between plants and soil microbes to community members sharing resources.

WEEK 4

20m General Science: Grade 5 - Lesson 4

Colony Collapse Disorder

Materials: The Hive Detectives

→ INTRO
What did you read last time? How do you think other beekeepers and scientists reacted when they heard what happened to Dave's bees?

→ READ, NARRATE, & DISCUSS
p.15-17 "In all my" - "and stop it."

• How do you think the scientists might test their three theories?

→ SUPPLEMENTAL
∞ Video Link: How Do Bees Make Honey? (to timestamp 3:20)

→ NOTE
Read A Dream Team of Bee Scientists p.18-19, if desired.

• FIELD TRIP
Visit a beekeeper. Notice the bees come and go from the hive. Notice how they interact with each other at the entrance. Notice their behavior when the beekeeper opens the hive. Can you find the queen? How does she look different? Where is she in the hive? Are there larvae in the hive? Where are they?

WEEK 4

30m Labs: Grade 5 - Lesson 4

Lab: What is Honey?

Materials: Grade 5 Lab Book and materials listed within

→ LAB DAY
Complete day 4 of What is Honey?, as directed in the Lab Book.

Suggested day 4: Students complete the Analysis and Conclusion today.

WEEK 5

15m Natural History: Grade 5 - Lesson 5

Plants in Community

Materials: The Forest in the Tree, Shanleya's Quest card game

PREP: Read Teacher Tip

→ INTRO
Recall what we learned about how plants work and how they are part of a community.

→ READ, NARRATE, & DISCUSS
p.2-15 "Water wakes me" - "'Glomus,' I agree."

• How are the plant and the fungi working together?
• Look at the picture on p.15. What happened when the fungi connected?

★ TEACHER TIP
Are your learner(s) demonstrating more familiarity/interest in how plants work? More interest/ease in describing/identifying different types of plants? If not, consider practicing together more often, exploring extra helpings, or taking a field trip to a botanical garden.

• OUTDOOR WORK
Weekly walks and daily

[Click THIS text or scan the QR code for links.]

Term 1

(Notice the trees are also connected.)
● Play "memory" or "slap flower'" based on student preference, if you have the Shanleya's Quest deck.

outdoor time are vital! Resources in Quick Links.

WEEK 5

⬚ **20m General Science: Grade 5 - Lesson 5**

Learning from Bees

▯ Materials: The Hive Detectives

PREP: Read Teacher Tip

➔ INTRO
What did you read last time? The first step to solving a problem is to learn more, so let's see how they do that today.

➔ READ, NARRATE, & DISCUSS
p.20-25 "The CCD Working Group" - "But what?"

● Describe the samples scientists collected.
● What did the scientists learn so far from studying the bees?
● How does a bee pollinate flowers? It's actually pretty amazing! Watch the videos to take a closer look:
∞ Video Link: The Buzz About Bees (2:46)
∞ Video Link: Sweet Pea Pollination (2:13)

➔ NOTE
Read Inside the Hive p.26-27, if desired.

★ TEACHER NOTE
The second supplemental video mentions evolution at 0:24. Teachers might choose to introduce the term by the way, mute the narration, or skip this one.

★ TEACHER TIP
Are your learner(s) demonstrating more familiarity/interest in bees? More interest/ease in describing or thinking about what material Things are made of? If not, consider observing together more often, exploring extra helpings, or taking a field trip to a nature preserve, bee keeper, or museum with a chemistry exhibit. Reach out through Contact Us, if you need help.

● NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

WEEK 5

⬚ **30m Labs: Grade 5 - Lesson 5**

Lab: Plant Food

▯ Materials: Grade 5 Lab Book and materials listed within

PREP: Read Teacher Tip

➔ LAB DAY
Complete day 1 of Plant Food, as directed in the Lab Book.

Suggested day 1: Students read the Introduction, compose the prelab narration, and complete steps 1-6 of the Procedure, as directed in the Lab Book. This is a set-up and wait lab, so students will move on to the next lab while waiting for their plants to grow.

★ TEACHER TIP
Be sure to have flowers for next week's lab!

Science: Grade 5

[Click THIS text or scan the QR code for links.]

Term 1

WEEK 6 ⬚ **15m Natural History: Grade 5 - Lesson 6**

The Life of a Plant

▯ Materials: The Forest in the Tree, Shanleya's Quest card game

→ INTRO
Recall what happened in the story last time. How do you think this community will live, grow, and change? Will they have new neighbors, new babies, illnesses, etc, just like our community? Do you think plant communities experience these parts of life? Why or why not?

→ READ, NARRATE, & DISCUSS
p.16-32 "Broma is older" - "shape the Earth."

• If you have the Shanleya's Quest deck, play the same game as last week or try "Crazy Flower" or "Wildflower Rummy."

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 6 ⬚ **20m General Science: Grade 5 - Lesson 6**

The Pest Hypothesis

▯ Materials: The Hive Detectives

PREP: Read Teacher Tip

→ INTRO
What did you read last time? One hypothesis is that a known pest is killing the bees. What's a hypothesis? (a really good guess based on what we know) To find out if the guess that a known pest is killing the bees is accurate, they have to test it. Let's find out how they do that today.

→ READ, NARRATE, & DISCUSS
p.29-31 "When asked" - "Strike three."

→ SUPPLEMENTAL
∞ Video Link: How Do Bees Make Honey? (timestamp 3:20 to end)

→ NOTE
Read Inside the Hive p.32-33, if desired.

★ TEACHER TIP
Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to the discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

WEEK 6 ⬚ **30m Labs: Grade 5 - Lesson 6**

Lab: Flowers are for Pollinators

▯ Materials: Grade 5 Lab Book and materials listed within

→ LAB DAY
Complete Flowers are for Pollinators, as directed in the Lab Book.

Science: Grade 5

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Term 1

WEEK 7

☐ **15m Natural History: Grade 5 - Lesson 7**

Caring for My Garden Activity

Materials: The Forest in the Tree, plantings from earlier in term, map/globe

→ OBSERVE
Spend a few minutes carefully observing your various seedlings. How has their appearance developed? Are you noticing distinctive characteristics yet?

→ NARRATE
Use a nature journal to record what you observed.

→ RECAP
What was the Forest in the Tree about? What kind of tree was Broma? (cacao) Where was Broma? Find the Amazon on the map/globe.

→ READ, NARRATE, & DISCUSS
p.37 "Theobroma cacao is" - "Ivory Coast to Ghana."

• Find Ivory Coast and Ghana. The cacao tree belongs to the mallow family, which means that the hibiscus flower and the linden tree are its cousins in places like Canada, the U.S., Finland, Korea, and New Zealand! Find these locations as time permits.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 7

☐ **20m General Science: Grade 5 - Lesson 7**

The Virus Hypothesis

Materials: The Hive Detectives

→ INTRO
What did you read last time? Another hypothesis is that a virus might be making the bees sick, but viruses can't be seen with our eyes or even a microscope. How do you think they will test this hypothesis?

→ READ, NARRATE, & DISCUSS
p.34-37 "Diana Cox-Foster" - "raised indoors?"

• How might Diana design her study so that she can find out what she needs to know without making the CCD problem worse?
• What do you think—can honeybees be raised indoors?

WEEK 7

☐ **30m Labs: Grade 5 - Lesson 7**

Lab: Separation Challenge

Materials: Grade 5 Lab Book and materials listed within

→ LAB DAY
Complete day 1 of Separation Challenge, as directed in the Lab Book.

Science: Grade 5

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Term 1

	Suggested day 1: Students read the Introduction, compose the prelab narration, and begin the Procedure today. They may break for the day wherever they are in the Procedure.	
WEEK 8	⬚ **15m Natural History: Grade 5 - Lesson 8** *Plants in Spring and Summer* ▯ Materials: Nature All Around: Plants, Shanleya's Quest card game **→ INTRO** Recall what we learned about how plants work. You may look back at the pictures to help. Have you noticed that different plants grow in different seasons? **→ READ, NARRATE, & DISCUSS** p.12-15 "One of the" - "they are absorbed." • What plants have you collected so far? Which families do they belong to? Do they grow and bloom primarily in the spring or summer? Where do they grow? • If you have the Shanleya's Quest deck, play your favorite game. Is it getting easier to recognize their characteristics?	**• OUTDOOR WORK** Weekly walks and daily outdoor time are vital! Resources in Quick Links.
WEEK 8	⬚ **20m General Science: Grade 5 - Lesson 8** *Thinking about Bee Health* ▯ Materials: The Hive Detectives **→ INTRO** What did you read last time? **→ READ, NARRATE, & DISCUSS** p.37-39 "Well, not yet." - "a greater issue?" **→ NOTE** Read Bee Bodies p.40-41, if desired.	**• NATURE NOTEBOOK** Prompt for General Science in Outdoor Work Quick Link.
WEEK 8	⬚ **30m Labs: Grade 5 - Lesson 8** *Lab: Separation Challenge* ▯ Materials: Grade 5 Lab Book and materials listed within **→ LAB DAY** Complete day 2 of Separation Challenge, as directed in the Lab Book. Suggested day 2: Students complete the Procedure, as well as Analysis and Conclusions today.	

Science: Grade 5

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Term 1

WEEK 9 ⬚ **15m Natural History: Grade 5 - Lesson 9**

Plants in Fall and Winter

Materials: Nature All Around: Plants, Shanleya's Quest card game

→ INTRO
Tell me what you remember about plants in the spring and summer. What happens to plants in the fall and winter? Let's read and learn something new.

→ READ, NARRATE, & DISCUSS
p.16-19 "By fall, many" - "replanted each year"

- What 3 observations do you think are most helpful to identify a plant?
- What plants have you collected so far? Do you know what happens to them in the fall and winter? How do they come back the next year? How do they propagate?
- If you have the Shanleya's Quest deck, play your favorite game or try a new one.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 9 ⬚ **20m General Science: Grade 5 - Lesson 9**

Caring for Creation

Materials: The Hive Detectives

→ INTRO
What did you read last time?

→ READ, NARRATE, & DISCUSS
p.43-47 "One year after" - "knows about bees."

- Tell what scientists now know about CCD and its causes.
- How does our knowing bees make a difference?

→ SUPPLEMENTAL
∞ Video Link: Can Mushrooms Save the Honeybee? (5:45)

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

WEEK 9 ⬚ **30m Labs: Grade 5 - Lesson 9**

Lab: Plant Food

Materials: Grade 5 Lab Book and materials listed within

→ LAB DAY
Complete day 2 of Plant Food, as directed in the Lab Book.

Suggested day 2: Students complete the Procedure, as well as Analysis and Conclusions.

Science: Grade 5

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WEEK 10 ⬚ **15m Natural History: Grade 5 - Lesson 10**

Garden Niches

Materials: Nature All Around: Plants, Shanleya's Quest card game

→ INTRO

What do you recall about plants in different seasons? What about different places? Do you think the plants are different in different parts of our country?

→ READ, NARRATE, & DISCUSS

p.20-23 "If you were" - "for next year."

• Find your zone on the map on p.20-21. Have you visited any other zones? Were the plants different in those other zones? Have you visited different habitats in your own or another zone? Describe some of the plants you remember. Were there characteristics or patterns that you remember? If you haven't noticed plants in another zone, you might enjoy seeing those at the link:
∞ Video Link: Wildflowers in Death Valley (3:57)
• Play a game with Shanleya's Quest, as time permits.

→ NOTE

Explore "More Strange Plants" on p.26-27 for interest, if desired.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 10 ⬚ **20m General Science: Grade 5 - Lesson 10**

Being Cared for By Creation

Materials: The Hive Detectives

→ INTRO

What did you read last time?

→ READ, NARRATE, & DISCUSS

p.49-52 "It's the middle" - "several local fairs.)"

• Let's think of some of the ways that people's lives are better because of the bees. Some ways might have been mentioned in the book, or might have nothing to do with the book.
• Make up a poem about bees together!

→ SUPPLEMENTAL

∞ Video Link: How to Plant a Pollinator Garden (3:21)
∞ Video Link: How to Harvest Honey (8:47)

WEEK 10 ⬚ **30m Labs: Grade 5 - Lesson 10**

Lab: Catch Up or...

Materials: Grade 5 Lab Book and materials listed within, image link

→ LAB DAY

Catch up on labs, plan a garden, or copy the linked bee diagram into your

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notebook:
∞ Image Link: Honey Bee Anatomy Diagram

WEEK 11

15m Natural History: Grade 5 - Lesson 11

Caring for My Garden Activity

Materials: plantings from earlier in term

PREP: Read Teacher Tip

→ INTRO
Your seedlings are probably much bigger now! Spend today carefully observing them. Do your observations reveal to which families they belong yet? If not, then keep observing and be patient!

→ OBSERVE

→ NARRATE & DISCUSS
Use a nature journal to record what you observed.

★ TEACHER TIP
When students take exams next week, evaluate what they recall, but more importantly, whether any of the new plants they have met are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc?

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 11

20m General Science: Grade 5 - Lesson 11

Final Thoughts

Materials: The Hive Detectives

PREP: Read Teacher Tip

→ INTRO
What did you read last time?

→ READ, NARRATE, & DISCUSS
p.53-55 "Mary repeats" - "see them through."

→ SUPPLEMENTAL
A master beekeeper shares some important thoughts about biodiversity in the linked article. Teachers might share some of those thoughts along with the video. What are some ways that we can help all the bees?
∞ Article Link: Honeybee Suite
∞ Video Link: This Bee Works 50 Times Harder (1:55)

→ NOTE
Read any parts of Appendix Bee p.56-59, if desired.

★ TEACHER TIP
When student(s) take exams next week, evaluate what they recall, but more importantly, whether bees, our dependence on them, or what material Things are made of are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less busyness, etc? Reach out through Contact Us, if you need help.

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

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WEEK 11	⬚ **30m Labs: Grade 5 - Lesson 11** *Lab: Catch Up or...* Materials: Grade 5 Lab Book and materials listed within, resource link **➔ LAB DAY** Use this time to catch up on lessons in this or another course, or plan a future pollinator-friendly garden. Simply choose a location and select a few native plants based on the conditions at the location. There are many native plant nurseries. Xerces is a great resource if you need help: ∞ Resource Link: Xerces Society	
WEEK 12	⬚ **15m Natural History: Grade 5 - Lesson 12** *Exam Week* **➔ EXAMS** Answer question(s) related to course. Questions will come from: The Forest in the Tree, Nature All Around: Plants	**• OUTDOOR WORK** Weekly walks and daily outdoor time are vital! Resources in Quick Links.
WEEK 12	⬚ **20m General Science: Grade 5 - Lesson 12** *Exam Week* **➔ EXAMS** Answer question(s) related to course. Questions will come from: The Hive Detectives	
WEEK 12	⬚ **30m Labs: Grade 5 - Lesson 12** *Lab: Exam Week* **➔ EXAMS** Answer question(s) related to course. Questions will come from: Grade 5 Lab Book	

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WEEK 1

⬚ **15m Natural History: Grade 5 - Lesson 13**

Working in the Dark

▯ Materials: The Bat Scientists

PREP: Read Teacher Tip

→ INTRO

Look at the picture on the cover. Have you watched bats flying at night before? What do you know about bats? Briefly look at p.8-10 based on interest before beginning.

→ READ & NARRATE

p.13-15 "Do you see" - "his favorite."

→ READ, NARRATE, & DISCUSS

p.15-16 "Merlin Tuttle" - "to save them."

→ SUPPLEMENTAL

Estimating the number of animals in a population, like bats in a cave, is challenging! Check out the linked photos and see if you can estimate the number of animals in just a glance.

∞ Image Link: Herd of Bison

∞ Image Link: School of Fish

∞ Image Link: Murmuration

★ TEACHER TIP

Sometimes more complex readings can be broken into smaller parts. In this passage, the first part introduces us to the bats and the second to the scientist who studies them.

★ TEACHER TIP

Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

● **OUTDOOR WORK**

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 1

⬚ **20m General Science: Grade 5 - Lesson 13**

Who What Where

▯ Materials: The Tornado Scientist

PREP: Read Teacher Tip

→ INTRO

What kinds of storms do you experience in your region? Many people have tornadoes, and this book is about how scientists are trying to better understand when and where these storms will occur. Why do you think that would be important? Our story takes place in the United States, but tornadoes don't pay attention to political boundaries! Canada is one of the most tornado-prone countries, and tornadoes occur all over the world.

→ READ, NARRATE, & DISCUSS

p.1-5 "Beep... beep...beep" - "ever wanted to be."

● Draw or tell about a storm that you remember.

★ TEACHER TIP

Severe weather can be scary for some students, and we want to honor their experiences. Notice how they respond to the story and respond with sensitivity and encouragement. Some will find the information helps them overcome their fears, while others may need more time.

● **NATURE NOTEBOOK**

Prompt for General Science in Outdoor Work Quick Link.

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Term 2

WEEK 1	⬚ **30m Labs: Grade 5 - Lesson 13** *Lab: Where Does Weather Come From? (Part 1)* Materials: Grade 5 Lab Book and materials listed within **➔ LAB DAY** Complete day 1 of Where Does Weather Come From? (Part 1), as directed in the Lab Book. Suggested day 1: Students read the Introduction and then compose the prelab narration in their lab notebook. These need not be more than 1-3 sentences at first, and teachers should feel free to scribe for students, as necessary. Spend remaining time gathering materials for next week. Any additional time can be used for diagramming, as suggested in the lab book.	★ TEACHER TIP Pacing is only a suggestion. Students should engage with the lab at a pace that is appropriate for their abilities and interests.
WEEK 2	⬚ **15m Natural History: Grade 5 - Lesson 14** *Watching Out for Bats* Materials: The Bat Scientists **➔ INTRO** What did we read in the last passage? Where are we? What is it like? Who are we with? Today, we'll read about bats in their ecosystem. **➔ READ, NARRATE, & DISCUSS** p.19-24 "The baking hot" - "fear of bats." (including p.21) • Use Google or a local resource to discover what the bats in your area eat and what eats them. Where do they live? Who else is dependent on them? • Discuss the food chain/web based on the reading OR for the bats in your area. Draw or diagram, if helpful.	**• OUTDOOR WORK** Weekly walks and daily outdoor time are vital! Resources in Quick Links.
WEEK 2	⬚ **20m General Science: Grade 5 - Lesson 14** *Robin's Story* Materials: The Tornado Scientist PREP: Read Teacher Tip **➔ INTRO** Recall what we read about in the last passage. Today, let's find out how Robin became a tornado scientist. **➔ READ, NARRATE, & DISCUSS** p.6-11 "Robin! Come and" - "family history, too." **➔ SUPPLEMENTAL** ∞ Video Link: 1986 Tornado Footage (4:58)	★ TEACHER TIP Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to the discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

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WEEK 2

☐ **30m Labs: Grade 5 - Lesson 14**

Lab: Where Does Weather Come From? (Part 1)

Materials: Grade 5 Lab Book and materials listed within

→ LAB DAY
Complete day 2 of Where Does Weather Come From? (Part 1), as directed in the Lab Book.

Suggested day 2: Students conduct the Procedure today. They may need help learning how to use the grid lines in their notebook to create a table for data collection, as shown in the lab book. Cut and paste, if needed. Today's procedure may take longer than the allotted time, so plan accordingly.

WEEK 3

☐ **15m Natural History: Grade 5 - Lesson 15**

Knowing Bats

Materials: The Bat Scientists

→ INTRO
Recall what we learned in the last passage. Let's get to know the bats a bit better in this passage.

→ READ, NARRATE, & DISCUSS
p.27-33 "Want to see" - "truth about bats."

● Discuss the wing diagram on p.33. What stands out to you? Copy the diagram, if desired.

● OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 3

☐ **20m General Science: Grade 5 - Lesson 15**

Twister Science

Materials: The Tornado Scientist

→ INTRO
Tell me what you remember from the last reading. Today, Robin is going to teach us about how a tornado works and how scientists learn about them.

→ READ, NARRATE, & DISCUSS
p.13-17 "Look at that!" - "of cold air."

→ SUPPLEMENTAL
∞ Video Link: What Causes a Tornado? (3:05)

WEEK 3

☐ **30m Labs: Grade 5 - Lesson 15**

Lab: Where Does Weather Come From? (Part 1)

Materials: Grade 5 Lab Book and materials listed within

→ LAB DAY
Complete day 3 of Where Does Weather Come From? (Part 1), as directed in the

Lab Book.

Suggested day 3: Students finish the lab today using the Analysis and Conclusions section. Support as needed to create another table using the grid lines in their notebook or by cutting and pasting. Then complete the postlab narration, as directed.

WEEK 4

15m Natural History: Grade 5 - Lesson 16

What's So Special About Bats?

Materials: The Bat Scientists

→ INTRO

Recall what we learned in the last passage. Can you think of any characteristics that make bats really special?

→ READ, NARRATE, & DISCUSS

p.34-39 "Bats are unique" - "second chance."

• Find the picture of a bat from your region or just a bat that you like at the link. Can you notice the adaptations discussed in today's reading? Does this tell you anything about this bat?

∞ Image Link: Bats by Region

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 4

20m General Science: Grade 5 - Lesson 16

Life Cycle of a Storm

Materials: The Tornado Scientist

PREP: Read Teacher Tip

→ INTRO

What do you remember about tornadoes from the last lesson? Let's hear more about the life cycle of these storms.

→ READ & NARRATE

p.18-19 "You probably know" - "spinning funnel."

→ READ, NARRATE, & DISCUSS

p.20-23 "Tornadoes are a" - "tornado chases you."

→ SUPPLEMENTAL

Today's reading discusses storm formation. Do you remember in what part of the atmosphere this happens? Check out the images at the following website. Can you find where weather occurs? Where weather balloons rise to? Where aurora borealis occurs?

∞ Website Link: Layers of the Atmosphere

★ TEACHER TIP

Some students may benefit from reading and narrating this lesson in parts, as presented here. Help them notice if there are parts to the reading; these can help them organize the ideas in their minds.

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WEEK 4

⬚ **30m Labs: Grade 5 - Lesson 16**

Lab: Meteorology

▯ Materials: Grade 5 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY

Complete day 1 of Meteorology, as directed in the Lab Book.

Suggested day 1: Students read the Introduction and prepare for data collection, which will take place at the same time every day for the next 1 month (or more, if they want). Then compose the prelab narration in their lab notebook. If they need some extra time to prepare for data collection, there is flexibility to push the schedule back 1 week.

★ TEACHER TIP

Setting an alarm or calendar reminder can facilitate data collection!

WEEK 5

⬚ **15m Natural History: Grade 5 - Lesson 17**

What Do You Do with a Problem?

▯ Materials: video link

→ INTRO

Recall what we learned in the last passage. Today, we have a video that struggles with a bat problem, so let's watch and see what we think.

→ VIEW & NARRATE

∞ Video Link: The Case for Vampire Bats (8:48)

- What do you think? What should be done for the people? What should be done for the bats?
- What makes a creature valuable?

→ NOTE

Explore "Batty Myths" on p.38 for interest, if desired.

★ TEACHER NOTE

The required video in this lesson includes a cartoon of a vampire bat biting a child at 0.18 and mentions evolution at 2:51. Teachers might allow these to pass by the way, but should preview if they have concerns. The video could be started at 0:27 if they think the cartoon will be frightening, and teachers could decide in advance how they want to address any questions about evolution.

● OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 5

⬚ **20m General Science: Grade 5 - Lesson 17**

Seeing Storms

▯ Materials: The Tornado Scientist

→ INTRO

Recall what you've read so far about storms. How do you think scientists see inside storms?

→ READ, NARRATE, & DISCUSS

p.24-29 "Spotting a tornado" - "storm is moving."

★ TEACHER NOTE

The section subtitled "Surviving the Greensburg Tornado" p.26-27 provides a detailed experience of a storm. Teachers should preread if they have any concerns. They might choose to skip this section or allow students to help

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		decide if they want to read it.
WEEK 5	⬚ **30m Labs: Grade 5 - Lesson 17** *Lab: Where Does Weather Come From? (Part 2)* Materials: Grade 5 Lab Book and materials listed within **➔ LAB DAY** Complete day 1 of Where Does Weather Come From? (Part 2), as directed in the Lab Book. Suggested day 1: Students read the Introduction and then compose the prelab narration in their lab notebook. Spend remaining time gathering materials for next week. Any additional time can be used for diagramming, as suggested in the lab book.	
WEEK 6	⬚ **15m Natural History: Grade 5 - Lesson 18** *Bat Habitat* Materials: The Bat Scientists PREP: Read Teacher Tip **➔ INTRO** Let's remember what our video was about and where we left off in our book. A lot of bats live in caves. Have you ever been to a cave? Why do you think bats like caves? **➔ READ, NARRATE, & DISCUSS** p.41-45 "You'd never know" - "hibernating bats like." p.45 "How a cave traps cold air." ● Describe the cave as a habitat. Copy the diagram, if desired.	★ TEACHER TIP Are your learner(s) demonstrating more familiarity/interest in bats or their role in the ecosystem? More interest/ease in describing/identifying different types of plants? If not, consider observing together more often, exploring extra helpings, or taking a field trip to a nature preserve or conservationist that has bats. **● OUTDOOR WORK** Weekly walks and daily outdoor time are vital! Resources in Quick Links.
WEEK 6	⬚ **20m General Science: Grade 5 - Lesson 18** *Weather Experiments* Materials: The Tornado Scientist PREP: Read Teacher Tip **➔ INTRO** Recall from the last reading how scientists "see" storms. What do they do with their observations?	★ TEACHER TIP Are your learner(s) demonstrating more familiarity/interest in weather? More interest/ease in collecting weather data or thinking about the instruments used to do so? If not, consider observing together more often, exploring extra helpings, or taking a field

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	→ **READ, NARRATE, & DISCUSS** p.30-33 "My job is" - "target? Dixie Alley."	trip to a weather station. Reach out through Contact Us, if you need help.
WEEK 6	⬚ **30m Labs: Grade 5 - Lesson 18** *Lab: Where Does Weather Come From? (Part 2)* Materials: Grade 5 Lab Book and materials listed within → **LAB DAY** Complete day 2 of Where Does Weather Come From? (Part 2), as directed in the Lab Book. Suggested day 2: Students conduct the Procedure today.	
WEEK 7	⬚ **15m Natural History: Grade 5 - Lesson 19** *Stewardship* Materials: The Bat Scientists PREP: Read Teacher Tip → **INTRO** Tell me what you learned about bats in the last reading. Knowing how important caves are, what do you think we can do to take care of them? → **READ, NARRATE, & DISCUSS** p.45-51 "People make" - "survive, too." ● The author uses the term "umbrella species." If you are familiar with the term "keystone species," what do you think is similar or different about the idea of "umbrella species"? The habitat is an important part of this idea - discuss the difference between using something, such as a plant, animal, or cave, without regard for its goodness versus responsible stewardship. → **NOTE** Explore "Gating Laurel Cave" on p.48-49 for interest, if desired.	★ TEACHER TIP Keystone species and umbrella species are sometimes used interchangeably, but an umbrella species usually migrates, has high habitat needs, and affects a great number of species also connected to its habitat over its entire range. ● **OUTDOOR WORK** Weekly walks and daily outdoor time are vital! Resources in Quick Links.
WEEK 7	⬚ **20m General Science: Grade 5 - Lesson 19** *Still Learning* Materials: The Tornado Scientist → **INTRO** Remind me what you've learned about how scientists "see" storms and what they do with those observations. So, do you think they've got tornadoes figured out? → **READ, NARRATE, & DISCUSS** p.35-41 "Eighth-grader Andrew" - "stop: Huntsville, Alabama."	★ TEACHER NOTE The first part of this reading p.35-36 provides a detailed experience of a storm. Teachers should preread if they have any concerns. They might choose to begin the reading from the subtitle "Southern Shocker" p.36 or allow students to help

	• Describe the characteristics that make southern tornadoes and plains tornadoes different.	decide if they want to read it.
WEEK 7	**30m Labs: Grade 5 - Lesson 19** *Lab: Where Does Weather Come From? (Part 2)* Materials: Grade 5 Lab Book and materials listed within **→ LAB DAY** Complete day 3 of Where Does Weather Come From? (Part 2), as directed in the Lab Book. Suggested day 3: Students finish the lab today using the Analysis and Conclusions section, completing the postlab narration, as directed.	
WEEK 8	**15m Natural History: Grade 5 - Lesson 20** *More Bat Habitats* Materials: The Bat Scientists PREP: Read Teacher Tip **→ INTRO** Tell me what you learned about bats in the last reading. There are a few other kinds of homes that bats like, and some of them might surprise you. **→ READ, NARRATE, & DISCUSS** p.53-58 "A woman gently" - "busy bridge." **→ NOTE** Explore "Celebrity Bridge Bats" on p.60-61 for interest, if desired. ∞ Video Link: Austin's Nocturnal Neighbors (1:27)	★ TEACHER TIP Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s). **• OUTDOOR WORK** Weekly walks and daily outdoor time are vital! Resources in Quick Links.
WEEK 8	**20m General Science: Grade 5 - Lesson 20** *The Tornado Trap* Materials: The Tornado Scientist **→ INTRO** Recall what is happening with Robin and her fellow weather scientists. What kinds of questions do you think they should ask about storms? How can they collect better observations? **→ READ, NARRATE, & DISCUSS** p.42-51 "Where do scientists" - "goes into SASSI."	

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[Click THIS text or scan the QR code for links.]

Term 2

• What information have you been recording on your weather chart? How is the data collected by the scientists in your book similar or different? Take a closer look at how they collect this data in their tornado trap:
∞ Video Link: Storm Pods (1:34)

WEEK 8

⬚ **30m Labs: Grade 5 - Lesson 20**

Lab: Meteorology

▯ Materials: Grade 5 Lab Book and materials listed within

→ LAB DAY
Complete Day 2 of Meteorology, as directed in the Lab Book.

Suggested Day 2: Students finish the lab today using the Analysis and Conclusions section. Students may need help completing their line graph for analysis. Then complete the postlab narration, as directed.

WEEK 9

⬚ **15m Natural History: Grade 5 - Lesson 21**

Stewarding Problems

▯ Materials: The Bat Scientists

→ INTRO
Tell me what you know about where bats live. Let's find out how learning more about their habitats can help us make better choices and do the right thing.

→ READ, NARRATE, & DISCUSS
p.62-64 "Though many" - "of last refuge."

• Make up a poem about bats together!

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 9

⬚ **20m General Science: Grade 5 - Lesson 21**

Problem-Solving

▯ Materials: The Tornado Scientist

PREP: Read Teacher Tip

→ INTRO
Tell me how the tornado research project is going so far. Let's find out how Robin is going to problem-solve.

→ READ, NARRATE, & DISCUSS
p.52-57 "Back in Alabama" - "Bingo. It's working."

→ SUPPLEMENTAL
Is the humble weather balloon still important?
∞ Video Link: Weather Balloons (2:43)

★ TEACHER TIP
Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to the discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

• NATURE NOTEBOOK
Prompt for General

Science: Grade 5

Click THIS text or scan the QR code for links.

Term 2

Science in Outdoor Work Quick Link.

WEEK 9

⬚ **30m Labs: Grade 5 - Lesson 21**

Lab: Robotics Activity

Materials: Grade 5 Lab Book and materials listed within

→ LAB DAY

Complete day 1 of Robotics Activity, as directed in the Lab Book.

Suggested day 1: Students read the Introduction and begin working with their robotics kit—no prelab narration for this activity. Students may draw, diagram, or dictate in their lab notebook, just as they would in their nature journals.

WEEK 10

⬚ **15m Natural History: Grade 5 - Lesson 22**

White Nose Syndrome

Materials: The Bat Scientists

→ INTRO

Tell me what you learned about bat habitat in the last reading. Bats have another challenge in addition to homelessness. It is called white nose syndrome, so let's read about that today.

→ READ, NARRATE, & DISCUSS

p.67-70 "After sleeping" - "our lifetimes."

→ NOTE

Explore "Cleanable Caves" on p.71 for interest, if desired.

★ TEACHER NOTE

Evolution is mentioned in the third sentence on p.68: "Everything from... is uncertain." Teachers might let it pass by the way, decide how they would like to discuss, or consider adjusting the text if reading aloud.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 10

⬚ **20m General Science: Grade 5 - Lesson 22**

Reasons for Research

Materials: The Tornado Scientist

→ INTRO

Remember back to the beginning of the book. Why do scientists want to learn more about tornadoes?

→ READ, NARRATE, & DISCUSS

p.58-65 "The weather seems" - "ever wanted to be."

- Diagram how tornadoes form.

→ SUPPLEMENTAL

∞ Video Link: Hello Kitty Goes to Space (3:46)

• FIELD TRIP

Visit a meteorologist or weather station.

Science: Grade 5

[Click THIS text or scan the QR code for links.]

Term 2

WEEK 10 ⬚ **30m Labs: Grade 5 - Lesson 22**

Lab: Robotics Activity

▯ Materials: Grade 5 Lab Book and materials listed within

→ LAB DAY

Complete day 2 of Robotics Activity, as directed in the Lab Book.

Suggested day 2: Students finish working with their robotics kit and complete the Analysis and Conclusion section in the lab book, finishing any remaining observations and completing the postlab narration.

WEEK 11 ⬚ **15m Natural History: Grade 5 - Lesson 23**

Listening to Bats

▯ Materials: The Bat Scientists

PREP: Read Teacher Tip

→ INTRO

Remind me what we read in the last passage. Let's finish the book by learning how the bat scientists observe the bats.

→ READ, NARRATE, & DISCUSS

p.72-73 "Back in" - "it came from."
p.75-77 "Having done" - "without bats."

→ SUPPLEMENTAL

∞ Video Link: Using a Bat Detector (3:15)
∞ Video Link: Filming Bats (4:22)

→ NOTE

Explore "Killer Wind" on p.74 for interest, if desired.

★ TEACHER TIP When students take exams next week, evaluate what they recall, but more importantly, whether bats or their role in the ecosystem are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc?

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 11 ⬚ **20m General Science: Grade 5 - Lesson 23**

Catch Up

PREP: Read Teacher Tip

→ CATCH UP

Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

★ TEACHER TIP
When student(s) take exams next week, evaluate what they recall, but more importantly, whether tornadoes, weather, or instrumentation are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less busyness, etc? Reach out

Term 2

		through Contact Us, if you need help.
WEEK 11	⬚ **30m Labs: Grade 5 - Lesson 23** *Lab: Catch Up* ▯ Materials: Grade 5 Lab Book and materials listed within **→ CATCH UP** Use this time to catch up on lessons in this or another course, or explore Extra Helpings.	
WEEK 12	⬚ **15m Natural History: Grade 5 - Lesson 24** *Exam Week* **→ EXAMS** Answer question(s) related to course. Questions will come from: The Bat Scientists	**• OUTDOOR WORK** Weekly walks and daily outdoor time are vital! Resources in Quick Links.
WEEK 12	⬚ **20m General Science: Grade 5 - Lesson 24** *Exam Week* **→ EXAMS** Answer question(s) related to course. Questions will come from: The Tornado Scientist	
WEEK 12	⬚ **30m Labs: Grade 5 - Lesson 24** *Lab: Exam Week* **→ EXAMS** Answer question(s) related to course. Questions will come from: Grade 5 Lab Book	

Science: Grade 5

[Click THIS text or scan the QR code for links.]

Term 3

WEEK 1

⬚ **15m Natural History: Grade 5 - Lesson 25**

What Happened to the Forest

Materials: Beetle Busters

PREP: Read Teacher Tip

→ INTRO
Look at the picture on the cover. What do you notice about this insect? In this book, we will learn about the delicate ecological balance between insect and plant life and what happens when it goes awry.

→ READ & NARRATE
p.4-6 "Ryan Zumpano" - "right thing to do?"

→ READ, NARRATE, & DISCUSS
p.7-8 "People who" - "hardwood forest."

• Tell about a favorite tree.

★ TEACHER TIP
Some students will benefit from reading and narrating this selection in two parts, as shown: the first introduces us to the forest, and the second to the problem.

★ TEACHER NOTE
Evolution is mentioned in the second paragraph on p.8. Teachers might let it pass by the way, decide how they would like to discuss, or consider adjusting the text if reading aloud.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 1

⬚ **20m General Science: Grade 5 - Lesson 25**

The Launch

Materials: Mission to Pluto

PREP: Read Teacher Tip

→ INTRO
Pluto is at the edge of our solar system. This is a story about scientists who study this distant object. Why do you suppose they would want to do that?

→ READ, NARRATE, & DISCUSS
p.1-6, 10 "The large meeting" - "So stay tuned."

→ SUPPLEMENTAL
∞ Video Link: New Horizons Launch (1:35)

→ NOTE
Explore The Journey to Pluto p.8-9 for interest, if desired.

★ TEACHER TIP
Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to the discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

WEEK 1

⬚ **30m Labs: Grade 5 - Lesson 25**

Lab: The Makings of a Telescope

Materials: Grade 5 Lab Book and materials listed within

PREP: Read Teacher Tip

★ TEACHER TIP
Pacing is only a suggestion. Students should engage with the lab at a pace that is appropriate for their abilities and interests.

Science: Grade 5

[Click THIS text or scan the QR code for links.]

Term 3

→ LAB DAY
Complete day 1 of The Makings of a Telescope, as directed in the Lab Book.

Suggested day 1: Students read the Introduction and then compose the prelab narration in their lab notebook. These need not be more than 1-3 sentences at first, and teachers should feel free to scribe for students, as necessary. Spend remaining time gathering materials for next week. Any additional time can be used for diagramming, as suggested in the lab book.

WEEK 2

15m Natural History: Grade 5 - Lesson 26
The Beetle

Materials: Beetle Busters

→ INTRO
Recall what has happened in the story. Today, we will learn more about the beetle that is causing the trouble.

→ READ, NARRATE, & DISCUSS
p.9-13 "The Asian longhorned" - "entire neighborhood."

• If cutting trees in one community would save the trees in your neighborhood, is it worth it? What if you lived in the community whose trees were being cut? What if it were your favorite tree?

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 2

20m General Science: Grade 5 - Lesson 26
The Path to a Space Mission

Materials: Mission to Pluto

→ INTRO
Recall what the launch party was like and why people were so excited. Today let's find out how we got to the launch.

→ READ, NARRATE, & DISCUSS
p.11-17, 20 "Three years after" - "pad was on."

→ NOTE
Explore any sidebars and Pluto Primer p.18-19 for interest, if desired.

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

WEEK 2

30m Labs: Grade 5 - Lesson 26
Lab: The Makings of a Telescope

Materials: Grade 5 Lab Book and materials listed within

→ LAB DAY
Complete day 2 of The Makings of a Telescope, as directed in the Lab Book.

Suggested day 2: Students finish working with their lenses and complete the Analysis and Conclusion section in the lab book, finishing any remaining observations and completing the postlab narration.

Science: Grade 5

Click THIS text or scan the QR code for links.

Term 3

WEEK 3

⬚ **15m Natural History: Grade 5 - Lesson 27**

A Life in Rings

▯ Materials: Beetle Busters

PREP: Read Teacher Tip

→ INTRO

Tell me what you remember about the beetle. Can they tell if the tree has beetles inside it? What do you think it would look like on the inside? Today we are going to read a bit about how these tree scientists look inside the trees using something called dendrochronology.

→ READ, NARRATE, & DISCUSS

p.13-14 "Lots of things" - "the tree itself."

→ SUPPLEMENTAL

∞ Website Link: Dendrochronology at Mesa Verde

★ TEACHER TIP

Dendrochronology comes from the Greek words meaning "tree" and "time."

★ TEACHER TIP

Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 3

⬚ **20m General Science: Grade 5 - Lesson 27**

Building a Spacecraft

▯ Materials: Mission to Pluto

PREP: Read Teacher Tip

→ INTRO

Recall how Pluto was discovered, and eventually, scientists decided to study it further. To accomplish their mission, they will need a spacecraft. What do you think are important characteristics of a spacecraft?

→ READ, NARRATE, & DISCUSS

p.21-25, 28 "Outer space may" - "engineers have done."

→ NOTE

Explore any sidebars and The Spacecraft p.26-27 for interest, if desired.

★ TEACHER TIP

This passage describes the design of New Horizons. Therefore, the narration is more likely to be points of that description.

Science: Grade 5

[Click THIS text or scan the QR code for links.]

Term 3

WEEK 3

⬚ **30m Labs: Grade 5 - Lesson 27**

Lab: Rocket Science

Materials: Grade 5 Lab Book and materials listed within

→ LAB DAY

Complete day 1 of Rocket Science, as directed in the Lab Book.

Suggested day 1: Students read the Introduction and then compose the prelab narration in their lab notebook. Spend remaining time gathering materials for next week. Any additional time can be used for diagramming, as suggested in the lab book.

WEEK 4

⬚ **15m Natural History: Grade 5 - Lesson 28**

How the Beetle Got Here

Materials: Beetle Busters, globe

→ INTRO

Recall what dendrochronology is. Where do you think this beetle came from, and how did it get to North America? Look at your globe and find China. Then trace with your finger a path to North America.

→ READ, NARRATE, & DISCUSS

p.15-20 "The story of" - "in the country."

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 4

⬚ **20m General Science: Grade 5 - Lesson 28**

Collecting Data

Materials: Mission to Pluto

PREP: Read Teacher Tip

→ INTRO

Tell me what was important about the design of New Horizons. Today we'll read about the kinds of information they want to collect with New Horizons and what launch day was like for the scientists who designed it.

→ READ & NARRATE

p.28-32 "Looking for ice" - "It's a great job."

→ READ, NARRATE, & DISCUSS

p.32-34 "At last" - "it got there."

→ NOTE

Explore any sidebars and Seeing Across the Spectrum p.31 for interest, if desired.

★ TEACHER TIP

The first part of this passage describes the instruments on New Horizons (and the corresponding data to be collected), while the second narrates launch day from the scientists' perspective.

Science: Grade 5

[Click THIS text or scan the QR code for links.]

Term 3

WEEK 4

⬚ **30m Labs: Grade 5 - Lesson 28**

Lab: Rocket Science

▯ Materials: Grade 5 Lab Book and materials listed within

→ LAB DAY

Complete day 2 of Rocket Science, as directed in the Lab Book.

Suggested day 2: Students conduct steps 1 and 2 of the Procedure today.

WEEK 5

⬚ **15m Natural History: Grade 5 - Lesson 29**

An Infestation

▯ Materials: Beetle Busters

PREP: Read Teacher Tip

→ INTRO

Tell me the story so far of how the beetle got to North America. What does it matter if this beetle is in North America?

→ READ, NARRATE, & DISCUSS

Ch.2 p.21-24 "In August 2008" - "we have to lose."

• Knowing what a healthy tree should look like helped the scientist notice a sick tree quickly. Look at the picture in the Tree ID section on p.24. Do you know the leaf, fruit, twig, and bark of a tree in your place? Describe what you know.

★ TEACHER TIP

Are your learners demonstrating more familiarity/interest in native/invasive species or their role in the ecosystem? More interest/ease in describing/identifying particular native/invasive species? If not, consider observing together more often, exploring extra helpings, or taking a field trip to a nature preserve.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 5

⬚ **20m General Science: Grade 5 - Lesson 29**

The Path to Pluto

▯ Materials: Mission to Pluto

PREP: Read Teacher Tip

→ INTRO

What do you remember about New Horizons? What do you think it was like being one of the scientists who designed the spacecraft? How do you think you fly a spacecraft across space?

→ READ, NARRATE, & DISCUSS

p.35-39 "The Mission" - "said Alan."

→ SUPPLEMENTAL

∞ Video Link: Tour of the Galaxy (2:59)

★ TEACHER TIP

Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to the discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

• NATURE NOTEBOOK

Prompt for General

Science: Grade 5

Click THIS text or scan the QR code for links.

Term 3

Science in Outdoor Work Quick Link.

WEEK 5

⬚ **30m Labs: Grade 5 - Lesson 29**

Lab: Rocket Science

Materials: Grade 5 Lab Book and materials listed within

→ LAB DAY

Complete day 3 of Rocket Science, as directed in the Lab Book.

Suggested day 3: Students complete the Procedure today, using the grid lines in their notebook to create a table for data collection, as shown in the lab book.

WEEK 6

⬚ **15m Natural History: Grade 5 - Lesson 30**

Working Together

Materials: Beetle Busters

→ INTRO

Remind me what we read about in the last passage. This sounds like a really big problem, so let's find out what they are going to do about it.

→ READ, NARRATE, & DISCUSS

p.25-30 "Clint McFarland" - "easier to bear."

• Explain how it helped them to first understand the beetle's life cycle.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 6

⬚ **20m General Science: Grade 5 - Lesson 30**

Planning for the Unexpected

Materials: Mission to Pluto

PREP: Read Teacher Tip

→ INTRO

Tell me about flying a spacecraft across space. What kinds of surprises do you think might come up on such a journey?

→ READ, NARRATE, & DISCUSS

p.39, 41-42, 46 "Practice makes perfect" - "Stay tuned."

→ SUPPLEMENTAL

∞ Video Link: Relive the Moon Landing (2:15)

→ NOTE

Explore Minimoons p.44-45 for interest, if desired.

★ TEACHER TIP

This is another description of points passage!

★ TEACHER NOTE

The supplemental reading "Minimoons" includes references to early planet/moon formation. Teachers might choose to allow it to pass by the way, take the opportunity to discuss other ideas, or skip this one.

Science: Grade 5

[Click THIS text or scan the QR code for links.]

Term 3

WEEK 6

⬚ **30m Labs: Grade 5 - Lesson 30**

Lab: Rocket Science

▯ Materials: Grade 5 Lab Book and materials listed within

➔ LAB DAY

Complete day 4 of Rocket Science, as directed in the Lab Book.

Suggested day 4: Students finish the Analysis and Conclusion section. Students may need help completing their scatter plot for analysis. Then complete the postlab narration, as directed.

WEEK 7

⬚ **15m Natural History: Grade 5 - Lesson 31**

Beetle Busting

▯ Materials: Beetle Busters

➔ INTRO

Recall how they decided to problem-solve the beetle infestation. Let's read more about the process today.

➔ READ, NARRATE, & DISCUSS

p.31-34 "Joining a tree" - "thirty years?"

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 7

⬚ **20m General Science: Grade 5 - Lesson 31**

What is a Planet Anyway?

▯ Materials: Mission to Pluto

PREP: Read Teacher Tip

➔ INTRO

What do you recall about what it's like to fly a spacecraft?

➔ READ, NARRATE, & DISCUSS

p.40-41 sidebar "As New Horizons" - "that Pluto fails."

- It seems that we humans like to classify and group Things. What groups do we use for planets? What other Things do you know that we classify into groups? Animals? Plants? Rocks?
- Act out a news flash to tell people why Pluto is no longer classified as a planet.

★ TEACHER TIP

Are your learner(s) demonstrating more familiarity/interest in the solar system? More interest/ease in observing outer space or the technology used to do so? If not, consider observing together more often, exploring extra helpings, or taking a field trip to a planetarium. Reach out through Contact Us, if you need help.

WEEK 7

⬚ **30m Labs: Grade 5 - Lesson 31**

Lab: Making a Model

▯ Materials: Grade 5 Lab Book and materials listed within

PREP: Read Teacher Tip

➔ LAB DAY

Complete day 1 of Making a Model, as directed in the Lab Book.

★ TEACHER TIP

Teachers should verify that students have what they need for next week.

Click THIS text or scan the QR code for links.

Term 3

Suggested day 1: Students read the Introduction and then compose the prelab narration in their lab notebook. They should spend remaining time gathering supplies and/or communicating to their teacher what supplies they need for their model.

WEEK 8

15m Natural History: Grade 5 - Lesson 32

Learning from the Forest

Materials: Beetle Busters

→ INTRO

Tell me what you remember from the last passage. The scientists didn't want to just solve the problem, though. They wanted to learn from it, and that's what we're going to read about today.

→ READ, NARRATE, & DISCUSS

p.35-40 "Early in" - "and beyond."

→ NOTE

Explore "More Beetle Busting" on p.41-42 for interest, if desired.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 8

20m General Science: Grade 5 - Lesson 32

Hello from Pluto!

Materials: Mission to Pluto

→ INTRO

Remind me: Is Pluto a planet or not? Why or why not? Let's continue there with New Horizons.

→ READ, NARRATE, & DISCUSS

p.47-49, 52-53 "It had already" - "are studied."

→ NOTE

Explore Pluto Encounter p.50-51 for interest, if desired.

★ TEACHER NOTE

The age of the Earth is mentioned in the first full paragraph on p.53: "for billions of years." Teachers might choose to allow it to pass by the way, decide how they would like to discuss with students, or substitute "for a very long time" if reading aloud.

WEEK 8

30m Labs: Grade 5 - Lesson 32

Lab: Making a Model

Materials: Grade 5 Lab Book and materials listed within

→ LAB DAY

Complete day 2 of Making a Model, as directed in the Lab Book.

Suggested day 2: Students conduct the Procedure today, doing most of the work on their model.

Science: Grade 5

[Click THIS text or scan the QR code for links.]

Term 3

WEEK 9

⬚ **15m Natural History: Grade 5 - Lesson 33**

Analyzing Data

| Materials: Beetle Busters |

→ INTRO

Recall what the scientists did to learn more about the beetle infestation. They found out three things from their special study, so let's find out what they learned.

→ READ, NARRATE, & DISCUSS

p.43-46 "Back in the" - "those forests."

● How do you think the knowledge they gained helps?

● OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 9

⬚ **20m General Science: Grade 5 - Lesson 33**

Pluto's Secrets

| Materials: Mission to Pluto |

PREP: Read Teacher Tip

→ INTRO

Tell me what happened in the last reading on the day of the flyby. Let's find out about some of the first insights that New Horizons has shared.

→ READ, NARRATE, & DISCUSS

p.53-57, 60 "An active world" - "incredible beauty."

● Make up a poem about Pluto together!

→ SUPPLEMENTAL

∞ Resource Link: NASA New Horizons

→ NOTE

Explore New Horizons' Discoveries So Far p.58-59 for interest, if desired.

★ TEACHER TIP

This is another description of points passage!

★ TEACHER NOTE

The early formation of the Earth is discussed in the last paragraph on p.54: "Earth's surface... newborn planet." Teachers might choose to let it pass by the way or consider alternative theories they would like to present.

WEEK 9

⬚ **30m Labs: Grade 5 - Lesson 33**

Lab: Making a Model

| Materials: Grade 5 Lab Book and materials listed within |

→ LAB DAY

Complete day 3 of Making a Model, as directed in the Lab Book.

Suggested day 3: Students complete any final work on their model and complete the Analysis and Conclusion section in the lab book.

Science: Grade 5

[Click THIS text or scan the QR code for links.]

Term 3

WEEK 10 ⬚ **15m Natural History: Grade 5 - Lesson 34**

Succession Takes a Long Time

Materials: Beetle Busters

PREP: Read Teacher Tip

→ INTRO
Remind me what the scientists found out from their work. Let's read about how they expect the forest to recover in a process called succession.

→ READ, NARRATE, & DISCUSS
p.47-52 "Change happens" - "urban forest."

• What does the word "succession" mean? How does it make sense to describe what's happening in this situation?
• If students are familiar with succession from Life on Surtsey, help them compare/contrast the primary succession on Surtsey and the secondary succession here.

★ TEACHER TIP
Notice that this reading is about the concept of succession and what that means in regard to the infested forest. Students may narrate the sequence of events of succession or of the author's thought, but they may instead give points to describe the concept and its relevance to the story.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 10 ⬚ **20m General Science: Grade 5 - Lesson 34**

Meet the Neighbors

Materials: Mission to Pluto

→ INTRO
Recall some of the discoveries made by New Horizons. So, there is this spacecraft flying out at the edge of our solar system. What do you think happens to it now that it has made it to Pluto?

→ READ, NARRATE, & DISCUSS
p.61-63 "What's next" - "go find them."

★ TEACHER NOTE
The early formation of the solar system is discussed in the first full paragraph on p.62: "KBOs also hold... John Spencer." Teachers might choose to let it pass by the way or consider alternative theories they would like to present.

WEEK 10 ⬚ **30m Labs: Grade 5 - Lesson 34**

Lab: Driving by Computer

Materials: Grade 5 Lab Book and materials listed within

→ LAB DAY
Complete the Driving by Computer Activity, as directed in the Lab Book.

Students read the Introduction and begin the Procedure with their partner—no prelab narration for this activity. When finished with the activity, students complete the Analysis and Conclusion section in the lab book.

Science: Grade 5

[Click THIS text or scan the QR code for links.]

Term 3

WEEK 11 ⬚ **15m Natural History: Grade 5 - Lesson 35**

Dialogue is Key

▯ Materials: Beetle Busters

PREP: Read Teacher Tip

→ INTRO

Tell me how scientists expect the infested forest to recover. How do you think the people who live there feel about having all those trees cut?

→ READ, NARRATE, & DISCUSS

p.55-58 "But to the" - "I believe in them."

- Explain both points of view about whether or not the trees should be cut.
- What do you think about the problem of whether or not to cut down the infested trees? How can they make a good decision?

★ TEACHER TIP
This is another passage in which students might access the concluding ideas better by giving points to describe the author's view rather than narrating as a sequence of events.

★ TEACHER TIP When students take exams next week, evaluate what they recall, but more importantly, whether a plant/animal or their role in the ecosystem is more familiar or has possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc?

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 11 ⬚ **20m General Science: Grade 5 - Lesson 35**

Going and going...

▯ Materials: Mission to Pluto

PREP: Read Teacher Tip

→ INTRO

Recall what the plan is for New Horizons now that it has made its observations at Pluto. Let's finish the story today.

→ READ, NARRATE, & DISCUSS

p.63-65, 68 "Astronomers put" - "-five or so years."

→ SUPPLEMENTAL

∞ Video Link: New Horizons Tribute (2:35)

→ NOTE

Explore The Kuiper Belt p.66-67 for interest, if desired.

★ TEACHER TIP
When student(s) take exams next week, evaluate what they recall, but more importantly, whether the solar system, stars, or related technology are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less busyness, etc? Reach out through Contact Us, if you need help.

• NATURE NOTEBOOK

Science: Grade 5

Click THIS text or scan the QR code for links.

Term 3

		Prompt for General Science in Outdoor Work Quick Link.
WEEK 11	⬚ **30m Labs: Grade 5 - Lesson 35** *Lab: Catch Up* ▯ Materials: Grade 5 Lab Book and materials listed within **→ CATCH UP** Use this time to catch up on lessons in this or another course, or explore Extra Helpings.	
WEEK 12	⬚ **15m Natural History: Grade 5 - Lesson 36** *Exam Week* **→ EXAMS** Answer question(s) related to course. Questions will come from: Beetle Busters	**• OUTDOOR WORK** Weekly walks and daily outdoor time are vital! Resources in Quick Links.
WEEK 12	⬚ **20m General Science: Grade 5 - Lesson 36** *Exam Week* **→ EXAMS** Answer question(s) related to course. Questions will come from: Mission to Pluto	
WEEK 12	⬚ **30m Labs: Grade 5 - Lesson 36** *Lab: Exam Week* **→ EXAMS** Answer question(s) related to course. Questions will come from: Grade 5 Lab Book	

Science: Grade 5

Examination

Term 1

General Science: Grade 5

· The Hive Detectives: How would a chemist answer the question “What is honey?” OR Give possible reasons that honey bee populations are at risk, and explain what you can do to help.

· Explain how you did one investigation in science lab this term and what you learned from it.

Natural History: Grade 5

· The Forest in the Tree & Nature All Around: Plants: Tell about or draw what you know about how plants work. OR Compare in word or picture three different plant types that you found this term.

Term 2

General Science: Grade 5

· The Tornado Scientist: Tell about or draw how meteorologists use instruments to conduct field work, like studying tornadoes. OR Explain what impact scientists hope to have by studying severe weather events, like tornadoes.

· Explain how you did one investigation in science lab this term and what you learned from it.

Natural History: Grade 5

· The Bat Scientists: Explain in words or pictures what bats need to thrive in an ecosystem. OR Tell about or draw three unique characteristics of the bat.

Term 3

General Science: Grade 5

· Mission to Pluto: Tell about or draw one way that scientists learn about our universe. OR Explain why scientists study far away Things, like Pluto.

· Explain how you did one investigation in science lab this term and what you learned from it.

Natural History: Grade 5

· Beetle Busters: Explain in words or pictures why an invasive species is bad for an ecosystem. OR Tell about or diagram one way that scientists learn about the health of a forest.